

B1 Flows, Fluctuations and Complexity — Model Solutions, Trinity 2015

Navier–Stokes for incompressible viscous flow under gravity:

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \frac{1}{\rho} \nabla p + g \hat{\mathbf{k}} = \nu \nabla^2 \mathbf{u}. \quad (1)$$

Question 1 — 2D velocity decomposition, vorticity, enstrophy

Problem. Taylor-expand a 2D velocity field about the origin, identify the four kinematic components (translation, divergence, rotation, two strains); discuss dye evolution and the role of molecular diffusion in 2D turbulence; derive the 2D vorticity equation; explain why enstrophy is dissipated.

(a) Taylor expansion and kinematic decomposition

Taylor-expand $u(x, y)$ and $v(x, y)$ to first order about the origin:

$$u = u_0 + \left(\frac{\partial u}{\partial x}\right)_0 x + \left(\frac{\partial u}{\partial y}\right)_0 y + \cdots, \quad v = v_0 + \left(\frac{\partial v}{\partial x}\right)_0 x + \left(\frac{\partial v}{\partial y}\right)_0 y + \cdots.$$

Write each derivative as the symmetric / antisymmetric combination:

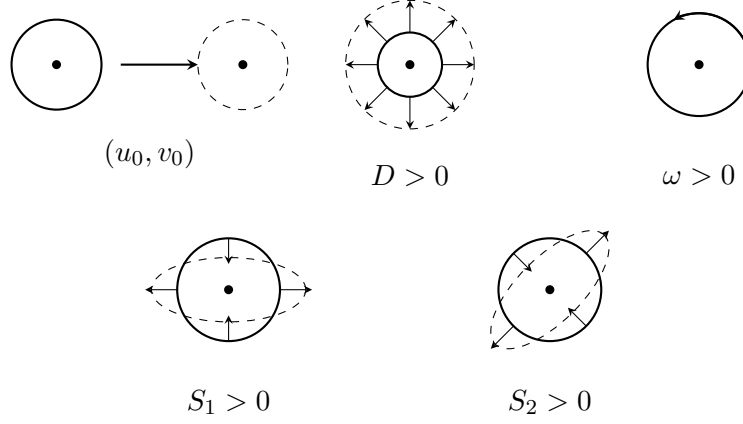
$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{1}{2}(D + S_1), & \frac{\partial v}{\partial y} &= \frac{1}{2}(D - S_1), \\ \frac{\partial u}{\partial y} &= \frac{1}{2}(S_2 - \omega), & \frac{\partial v}{\partial x} &= \frac{1}{2}(S_2 + \omega). \end{aligned}$$

Substitute into the expansion:

$$\boxed{u = u_0 + \frac{D}{2}x - \frac{\omega}{2}y + \frac{S_1}{2}x + \frac{S_2}{2}y, \quad v = v_0 + \frac{D}{2}y + \frac{\omega}{2}x - \frac{S_1}{2}y + \frac{S_2}{2}x.} \quad (2)$$

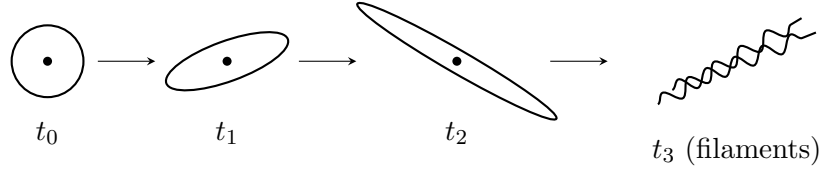
Physical interpretation:

- (u_0, v_0) — uniform translation of the parcel.
- $D = \nabla \cdot \mathbf{u}$ — isotropic dilation (expansion if $D > 0$, contraction if $D < 0$).
- ω — rigid-body rotation about the origin at angular rate $\omega/2$.
- S_1 — pure strain along the x -axis (stretches x , compresses y).
- S_2 — pure strain along the 45° axis (stretches along $y = x$, compresses along $y = -x$).



(b) Dye evolution in 2D incompressible turbulence

Incompressibility: $D = 0$, so the parcel area is preserved. Local flow is rotation + pure strain. Strain stretches the parcel exponentially along one principal axis and compresses it along the perpendicular axis at the same rate (area-preserving); rotation reorients the principal axes as the parcel is advected through the random strain field.



The parcel becomes an exponentially thinning, lengthening filament. Once the filament thickness reaches the diffusive cutoff $\ell_d \sim \sqrt{\kappa\tau}$ (where κ is the molecular diffusivity and τ the strain time), the dye gradient $\nabla c \sim c/\ell_d$ becomes large enough that molecular diffusion $\kappa\nabla^2 c$ smooths the field at the same rate as straining produces gradients — diffusion cannot be neglected, and dye is mixed irreversibly even though the underlying flow is reversible.

(c) 2D vorticity equation

Take the curl of (1) (gravity drops, $\nabla \cdot \mathbf{u} = 0$). For 2D flow $\mathbf{u} = (u, v, 0)$, vorticity $\boldsymbol{\omega} = \omega \hat{\mathbf{z}}$ with $\omega = \partial_x v - \partial_y u$. Apply ∂_y to the x -NS, ∂_x to the y -NS, subtract:

$$\frac{\partial}{\partial t}(\partial_x v - \partial_y u) + \partial_x[(u \cdot \nabla)v] - \partial_y[(u \cdot \nabla)u] = \nu \nabla^2(\partial_x v - \partial_y u).$$

Expand the advective terms:

$$\partial_x[u\partial_x v + v\partial_y v] - \partial_y[u\partial_x u + v\partial_y u].$$

Use $\partial_x u + \partial_y v = 0$ to cancel cross terms:

$$= u\partial_x \omega + v\partial_y \omega.$$

Collect:

$$\boxed{\left(\frac{\partial}{\partial t} + u\frac{\partial}{\partial x} + v\frac{\partial}{\partial y}\right)\omega = \nu \nabla^2 \omega.} \quad (3)$$

High-Re limit ($\nu \rightarrow 0$):

$$\boxed{\frac{D\omega}{Dt} = 0.} \quad (4)$$

Vorticity is materially conserved; in 2D there is no vortex stretching.

(d) Enstrophy is always dissipated

Multiply (3) by ω , integrate over the (unbounded or no-flux) domain. Advective term integrates to zero by incompressibility:

$$\frac{\partial}{\partial t} \iint \frac{\omega^2}{2} dx dy = \nu \iint \omega \nabla^2 \omega dx dy.$$

Integrate by parts (boundary terms vanish):

$$= -\nu \iint |\nabla \omega|^2 dx dy \leq 0.$$

Stretching strains thin vorticity filaments, increasing $|\nabla \omega|$, so the dissipation rate grows even as $\nu \rightarrow 0$. Hence enstrophy decays monotonically in 2D turbulence:

$$\boxed{\frac{\partial}{\partial t} \iint \frac{\omega^2}{2} dx dy = -\nu \iint |\nabla \omega|^2 dx dy \leq 0.}$$

Question 2 — Sound waves, Riemann invariants, shocks

Problem. From 1D adiabatic compressible Euler-with-viscosity, linearise to obtain $c = \sqrt{\partial p / \partial \rho}$; recast nonlinear inviscid form as Riemann invariants on characteristics; write the implicit simple-wave solution, sketch wave steepening, and estimate the shock thickness $\delta \sim \nu / c$.

(a) Linearisation and sound speed

Inviscid limit: $\nu = 0$. Perturb about rest:

$$u = u', \quad \rho = \rho_0 + \rho', \quad p = p_0 + p', \quad \rho', p', u' \ll .$$

Linearise momentum (drop $u \partial_x u$):

$$\frac{\partial u'}{\partial t} + \frac{1}{\rho_0} \frac{\partial p'}{\partial x} = 0.$$

Linearise continuity:

$$\frac{\partial \rho'}{\partial t} + \rho_0 \frac{\partial u'}{\partial x} = 0.$$

Adiabatic state law: $p = p_0(\rho/\rho_0)^\gamma$. Differentiate:

$$\frac{\partial p}{\partial \rho} = \gamma p_0 \rho_0^{-\gamma} \rho^{\gamma-1},$$

At rest:

$$\left. \frac{\partial p}{\partial \rho} \right|_0 = \frac{\gamma p_0}{\rho_0}.$$

So $p' = (\gamma p_0 / \rho_0) \rho'$. Eliminate ρ' :

$$\frac{\partial^2 u'}{\partial t^2} - \frac{\gamma p_0}{\rho_0} \frac{\partial^2 u'}{\partial x^2} = 0.$$

Wave equation, speed:

$$\boxed{c = \sqrt{\frac{\partial p}{\partial \rho}} = \sqrt{\frac{\gamma p_0}{\rho_0}}.} \quad (5)$$

Numerical estimate at sea level: $\gamma = 7/5$, $p_0 = 10^5$ Pa, $\rho_0 = 1.2$ kg m⁻³:

$$\gamma p_0 = 1.4 \times 10^5 = 1.4 \times 10^5 \text{ Pa.}$$

$$\frac{\gamma p_0}{\rho_0} = \frac{1.4 \times 10^5}{1.2} = 1.17 \times 10^5 \text{ m}^2 \text{ s}^{-2}.$$

$$c = \sqrt{1.17 \times 10^5} \approx 342 \text{ m s}^{-1}.$$

$$\boxed{c \approx 340 \text{ m s}^{-1}.}$$

(b) Riemann invariants

Write momentum (inviscid) and continuity:

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + \frac{1}{\rho} \frac{\partial p}{\partial x} = 0, \quad \frac{\partial \rho}{\partial t} + u \frac{\partial \rho}{\partial x} + \rho \frac{\partial u}{\partial x} = 0.$$

Use $p = p(\rho)$, so $\partial_x p = c^2 \partial_x \rho$, and $dP = dp/(\rho c)$ implies $\partial_x P = (c/\rho) \partial_x \rho$ and $\partial_t P = (c/\rho) \partial_t \rho$. Rewrite continuity:

$$\frac{\partial P}{\partial t} + u \frac{\partial P}{\partial x} + c \frac{\partial u}{\partial x} = 0.$$

Rewrite momentum (using $\partial_x p/\rho = c \partial_x P$):

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + c \frac{\partial P}{\partial x} = 0.$$

Add the two:

$$\frac{\partial}{\partial t}(u + P) + (u + c) \frac{\partial}{\partial x}(u + P) = 0.$$

Subtract:

$$\frac{\partial}{\partial t}(u - P) + (u - c) \frac{\partial}{\partial x}(u - P) = 0.$$

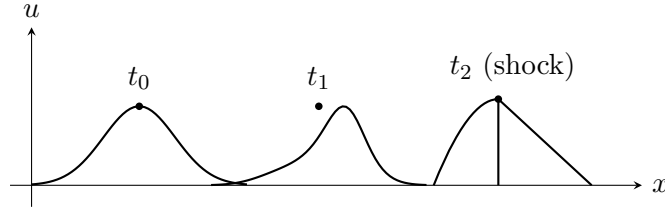
$$\boxed{\left[\frac{\partial}{\partial t} + (u + c) \frac{\partial}{\partial x} \right] (u + P) = 0, \quad \left[\frac{\partial}{\partial t} + (u - c) \frac{\partial}{\partial x} \right] (u - P) = 0.} \quad (6)$$

(c) Simple wave, shock formation, shock thickness

For a wave moving in $+x$ relative to mean flow, take $u - P = \text{const}$ everywhere (uniform upstream); then $P = u + \text{const}$, so $u + P = 2u + \text{const}$ is constant on C_+ characteristics $dx/dt = u + c$. Hence u itself is constant along C_+ :

$$\boxed{u(x, t) = u_0(x - (u + c)t),} \quad (7)$$

implicit because the characteristic speed depends on u .



Crests (large u) travel faster than troughs ($u + c$ depends on u); the wave leans forward and steepens until $\partial_x u \rightarrow -\infty$. Inviscid theory gives a multi-valued solution, unphysical: viscosity smooths the front, producing a thin shock.

Shock-thickness estimate: in the shock, balance the inertial term $u\partial_x u$ against the viscous term $\nu\partial_x^2 u$. Take $u \sim c$ (transonic) and $\partial_x \sim 1/\delta$:

$$u \frac{\partial u}{\partial x} \sim \frac{c^2}{\delta},$$

$$\nu \frac{\partial^2 u}{\partial x^2} \sim \frac{\nu c}{\delta^2}.$$

Equate:

$$\frac{c^2}{\delta} \sim \frac{\nu c}{\delta^2}.$$

Solve:

$$\boxed{\delta \sim \frac{\nu}{c}.} \quad (8)$$

Numerical estimate at sea level: $\nu = 1.5 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$, $c = 340 \text{ m s}^{-1}$:

$$\delta \sim \frac{1.5 \times 10^{-5}}{340}.$$

$$\delta \sim 4.4 \times 10^{-8} \text{ m}.$$

$$\boxed{\delta \sim 4 \times 10^{-8} \text{ m} \sim 100 \text{ atomic spacings}.}$$

Comparable to molecular mean free path; continuum description marginal.

Question 3 — Phase-space dynamics and a strange attractor

Problem. For $\dot{x} = yz$, $\dot{y} = x - y$, $\dot{z} = 1 - xy$: locate fixed points and prove instability; derive the phase-space continuity equation and show $\delta V = \delta V_0 e^{-(t-t_0)}$; show $\|\delta \mathbf{x}\| = \|\delta \mathbf{x}\|_0 e^{\lambda t}$ along an eigendirection of $(J + J^T)/2$; compute those eigenvalues at the origin; discuss the attractor's geometry.

(a) Fixed points and stability

Fixed points: $\dot{x} = \dot{y} = \dot{z} = 0$:

$$yz = 0, \quad x = y, \quad xy = 1.$$

$xy = 1$ rules out $y = 0$, so $z = 0$ and $x = y = \pm 1$:

$$\boxed{\mathbf{x}_{\pm} = (\pm 1, \pm 1, 0).}$$

Jacobian:

$$J = \begin{pmatrix} 0 & z & y \\ 1 & -1 & 0 \\ -y & -x & 0 \end{pmatrix}.$$

At $\mathbf{x}_+ = (1, 1, 0)$:

$$J_+ = \begin{pmatrix} 0 & 0 & 1 \\ 1 & -1 & 0 \\ -1 & -1 & 0 \end{pmatrix}.$$

Characteristic polynomial $\det(J_+ - rI) = 0$. Compute:

$$\det \begin{pmatrix} -r & 0 & 1 \\ 1 & -1-r & 0 \\ -1 & -1 & -r \end{pmatrix}.$$

Expand along the top row:

$$-r[(-1-r)(-r) - 0] + 1 \cdot [1 \cdot (-1) - (-1-r)(-1)].$$

Simplify:

$$\begin{aligned} & -r[r(1+r)] + [-1 - (1+r)]. \\ & -r^3 - r^2 - 2 - r. \end{aligned}$$

Multiply by -1 to standard form $r^3 + Ar^2 + Br + C = 0$:

$$r^3 + r^2 + r + 2 = 0.$$

Read off: $A = 1$, $B = 1$, $C = 2$. All positive. Test $C > AB$:

$$2 > 1 \cdot 1 = 1 \quad \checkmark.$$

By the stated criterion, $\mathbf{x}_+$ has at least one root with positive real part.

By the symmetry $(x, y) \rightarrow (-x, -y)$, $z \rightarrow z$ (which leaves the system invariant), $\mathbf{x}_-$ has the same characteristic polynomial. *Both fixed points are unstable.*

(b) Phase-space continuity and volume contraction

For $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$, the phase-space density $n(\mathbf{x}, t)$ obeys:

$$\frac{\partial n}{\partial t} + \nabla \cdot (n\mathbf{f}) = 0.$$

Apply to a comoving volume element δV containing fixed phase points. Conservation of points:

$$n \delta V = \text{const} \Rightarrow \frac{1}{\delta V} \frac{D\delta V}{Dt} = -\frac{1}{n} \frac{Dn}{Dt}.$$

Use the continuity equation along a streamline:

$$\frac{Dn}{Dt} = -n \nabla \cdot \mathbf{f}.$$

Combine:

$$\frac{D\delta V}{Dt} = (\nabla \cdot \mathbf{f}) \delta V.$$

Compute the divergence:

$$\nabla \cdot \mathbf{f} = \frac{\partial}{\partial x}(yz) + \frac{\partial}{\partial y}(x - y) + \frac{\partial}{\partial z}(1 - xy) = 0 - 1 + 0 = -1.$$

Solve:

$$\boxed{\delta V(t) = \delta V_0 e^{-(t-t_0)}}. \quad (9)$$

Volume contracts at constant rate. Trajectories are bounded (numerical evidence) yet repelled by both fixed points and not converging to one — so they collapse onto a zero-volume invariant set. That set is either a one-dimensional limit cycle or, if local stretching coexists with global contraction, a fractal-dimensional *strange attractor*.

(c) Symmetric-Jacobian growth rate and eigenvalues at origin

Linearise: $\delta \dot{\mathbf{x}} = J \delta \mathbf{x}$. Squared length:

$$\frac{d}{dt} \|\delta \mathbf{x}\|^2 = 2 \delta \mathbf{x} \cdot J \delta \mathbf{x}.$$

Symmetrise the inner product:

$$2 \delta \mathbf{x} \cdot J \delta \mathbf{x} = \delta \mathbf{x} \cdot (J + J^T) \delta \mathbf{x}.$$

Choose $\delta \mathbf{x}$ aligned with eigenvector of $(J + J^T)/2$ with eigenvalue λ :

$$\frac{d}{dt} \|\delta \mathbf{x}\|^2 = 2\lambda \|\delta \mathbf{x}\|^2.$$

Integrate (treating J as locally constant for short t):

$$\boxed{\|\delta \mathbf{x}\| = \|\delta \mathbf{x}\|_0 e^{\lambda(t-t_0)}}. \quad (10)$$

At the origin, $J(0, 0, 0) = \begin{pmatrix} 0 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$:

$$\frac{J + J^T}{2} = \begin{pmatrix} 0 & 1/2 & 0 \\ 1/2 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

z -direction decouples: $\lambda_3 = 0$. For the (x, y) block:

$$\det \begin{pmatrix} -\lambda & 1/2 \\ 1/2 & -1 - \lambda \end{pmatrix} = 0.$$

Expand:

$$\lambda(1 + \lambda) - 1/4 = 0.$$

$$\lambda^2 + \lambda - 1/4 = 0.$$

Quadratic formula:

$$\lambda = \frac{-1 \pm \sqrt{1+1}}{2}.$$

$$\boxed{\lambda \in \left\{ \frac{-1+\sqrt{2}}{2}, \frac{-1-\sqrt{2}}{2}, 0 \right\} \approx \{+0.207, -1.207, 0\}}.$$

(d) Geometry of the attractor

Discussion. Fractal dimension $D_f \approx 2.174$ lies between 2 and 3: the attractor is a sheet-like object with a fine, self-similar transverse structure — consistent with continual local stretching plus folding.

The volume-contraction result (9) shows phase-space volumes shrink at rate 1, so the sum of Lyapunov exponents is $\sum \lambda_i = -1$. From (10) the symmetric-part eigenvalues at the origin are $+0.207, 0, -1.207$; their sum is -1 , matching the trace. Local stretching ($\lambda_+ > 0$) coexists with strong contraction ($\lambda_- < 0$), the hallmark of a strange attractor: nearby trajectories diverge along the λ_+ axis while the volume collapses along λ_- , producing a fractal sheet. The two unstable fixed points $\mathbf{x}_\pm$ act as scrolling centres: the trajectory loops around one, ejects, and is captured by the other, building the two-lobed structure visible in the figure (qualitatively similar to a Lorenz-like attractor).

The Kaplan–Yorke estimate $D_f = 2 + \lambda_+ / |\lambda_-|$ gives, with the symmetric-Jacobian numbers, $D_f \approx 2 + 0.207/1.207 \approx 2.17$, consistent with the quoted 2.174 (true Lyapunov exponents would be needed for an exact match, but the agreement supports the strange-attractor identification).

Question 4 — Freely-jointed chains, loop closure, DNA looping

Problem. Find the most-probable end-to-end distance for an unconfined Gaussian FJC and for one tethered to a wall; obtain $S(z)$. Compute the loop-closure entropy $\Delta S = B - \frac{3}{2}k_B \ln N$; combine with the elastic loop energy to obtain the DNA loop free energy and find the optimal loop size; choose between Lac-repressor sites at 92 and 401 bp.

(a) Most-probable end-to-end distance, untethered FJC

Joint density:

$$P(\mathbf{r}) = \left(\frac{3}{2\pi N b^2} \right)^{3/2} \exp\left(-\frac{3r^2}{2N b^2} \right).$$

Radial distribution:

$$p(r) = 4\pi r^2 P(\mathbf{r}).$$

Maximise:

$$\frac{d}{dr} \left[r^2 \exp\left(-\frac{3r^2}{2N b^2} \right) \right] = 0.$$

Differentiate:

$$2r - r^2 \frac{3r}{N b^2} = 0.$$

Solve:

$$r_{\text{mp}}^2 = \frac{2N b^2}{3}.$$

$$\boxed{r_{\text{mp}} = b \sqrt{\frac{2N}{3}}.} \tag{11}$$

(b) Tethered FJC: most-probable distance and entropy

Joint density:

$$P(\mathbf{r}) = P_x(x)P_y(y) \frac{z}{\Omega} \exp\left(-\frac{3z^2}{2N b^2} \right), \quad z > 0.$$

Maximise $P(\mathbf{r})$ over (x, y, z) directly (not the radial form: the wall breaks isotropy). Gaussian factors P_x, P_y peak at $x = y = 0$:

$$x_{\text{mp}} = 0, \quad y_{\text{mp}} = 0.$$

Maximise $z \exp(-3z^2/2Nb^2)$:

$$\frac{d}{dz} \left[z \exp\left(-\frac{3z^2}{2Nb^2}\right) \right] = 0.$$

Differentiate:

$$1 - \frac{3z^2}{Nb^2} = 0.$$

Solve:

$$z_{\text{mp}} = b\sqrt{\frac{N}{3}}.$$

At the maximum $r = z$:

$$\boxed{r_{\text{mp}}^{\text{tethered}} = b\sqrt{\frac{N}{3}}.} \quad (12)$$

Comment. Smaller than (11) by $\sqrt{2}$: the wall removes the radial $4\pi r^2$ phase-space factor (which favoured larger r for the free chain) and replaces it with the linear z factor (only one direction available), so the most probable separation is reduced.

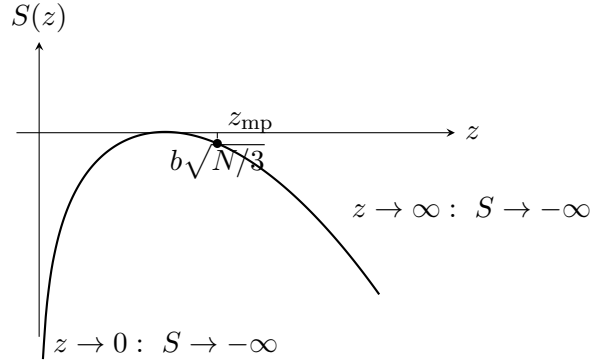
Entropy along z :

$$S(z) = k_B \ln P(z).$$

Substitute P_z :

$$S(z) = k_B \ln\left(\frac{z}{\Omega}\right) - \frac{3k_B z^2}{2Nb^2}.$$

$$\boxed{S(z) = k_B \ln\left(\frac{z}{\Omega}\right) - \frac{3k_B z^2}{2Nb^2}.} \quad (13)$$



Comment. $S \rightarrow -\infty$ as $z \rightarrow 0$ (only one configuration: chain crushed against wall) and as $z \rightarrow \infty$ (rare large excursion of a Gaussian). Maximum at $z_{\text{mp}} = b\sqrt{N/3}$, agreeing with the most-probable-distance calculation above. Stretching the chain away from z_{mp} reduces entropy and produces a restoring elastic force — the entropic origin of polymer elasticity.

(c) Loop-closure entropy

Density of $r = |\mathbf{r}|$ for the free chain:

$$p(r) = 4\pi r^2 \left(\frac{3}{2\pi N b^2} \right)^{3/2} \exp\left(-\frac{3r^2}{2N b^2}\right).$$

Limit $\delta \ll b$ (so $\delta \ll b\sqrt{N}$): exponential ≈ 1 :

$$\Pi(\delta) \approx 4\pi \left(\frac{3}{2\pi N b^2} \right)^{3/2} \int_0^\delta r^2 dr.$$

Integrate:

$$\Pi(\delta) \approx \frac{4\pi\delta^3}{3} \left(\frac{3}{2\pi N b^2} \right)^{3/2}.$$

$$\boxed{\Pi(\delta) = \frac{4\pi}{3} \left(\frac{3}{2\pi} \right)^{3/2} \left(\frac{\delta}{b} \right)^3 N^{-3/2}.} \quad (14)$$

Joining the ends restricts the configurational space from "all $\mathbf{r}$ " to " $|\mathbf{r}| < \delta$ ". The fractional volume retained is $\Pi(\delta)$. Boltzmann's $S = k_B \ln \Omega$ for fewer accessible states gives $\Delta S = k_B \ln \Pi(\delta)$ (entropy loss, $\Delta S < 0$).

Take the log:

$$\Delta S = k_B \ln \left[\frac{4\pi}{3} \left(\frac{3}{2\pi} \right)^{3/2} \left(\frac{\delta}{b} \right)^3 \right] - \frac{3}{2} k_B \ln N.$$

$$\boxed{\Delta S = B - \frac{3}{2} k_B \ln N, \quad B = k_B \ln \left[\frac{4\pi}{3} \left(\frac{3}{2\pi} \right)^{3/2} \left(\frac{\delta}{b} \right)^3 \right].} \quad (15)$$

(d) DNA loop free energy and Lac-repressor choice

A DNA segment of n base pairs has contour length $L_c = nL$ with $L = 0.34$ nm, and as an FJC contains $N = L_c/b = nL/b$ Kuhn segments. A circular loop of circumference L_c has radius:

$$R = \frac{L_c}{2\pi} = \frac{nL}{2\pi}.$$

Free energy of loop formation:

$$\Delta F = E_{\text{loop}} - T \Delta S.$$

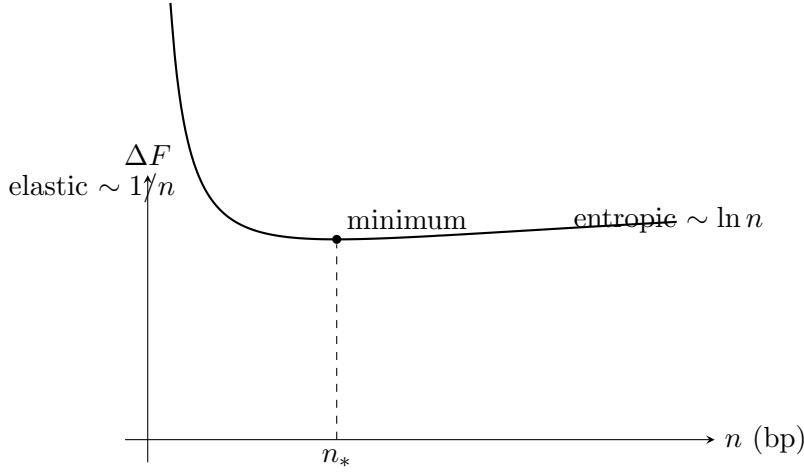
Substitute $E_{\text{loop}} = \pi k_B T b / (2R)$ and ΔS from (15):

$$\Delta F = \frac{\pi k_B T b}{2R} + \frac{3}{2} k_B T \ln N - T B.$$

Insert $R = nL/(2\pi)$ and $N = nL/b$:

$$\Delta F(n) = \frac{\pi^2 k_B T b}{nL} + \frac{3}{2} k_B T \ln \left(\frac{nL}{b} \right) - T B.$$

$$\boxed{\Delta F(n) = \frac{\pi^2 k_B T b}{nL} + \frac{3}{2} k_B T \ln \left(\frac{nL}{b} \right) + \text{const.}} \quad (16)$$



Minimise ΔF :

$$\frac{d\Delta F}{dn} = -\frac{\pi^2 k_B T b}{n^2 L} + \frac{3k_B T}{2n} = 0.$$

Solve:

$$\begin{aligned} \frac{3}{2n} &= \frac{\pi^2 b}{n^2 L} \\ n_* &= \frac{2\pi^2 b}{3L}. \end{aligned}$$

Insert $b = 100 \text{ nm}$, $L = 0.34 \text{ nm}$:

$$n_* = \frac{2\pi^2 \cdot 100}{3 \cdot 0.34}.$$

$$2\pi^2 = 19.74.$$

$$\frac{19.74 \cdot 100}{1.02} = 1935.$$

$$\boxed{n_* \approx 1.9 \times 10^3 \text{ bp.}} \quad (17)$$

Lac-repressor candidates: $n_1 = 92 \text{ bp}$, $n_2 = 401 \text{ bp}$. Both satisfy $n < n_*$, the regime where ΔF is dominated by the elastic $\propto 1/n$ term and *decreases* as n grows toward n_* . Hence:

$$\Delta F(401) < \Delta F(92).$$

$$\boxed{\text{The 401-bp pair is more likely to be linked by the Lac repressor.}} \quad (18)$$

(Quantitative comparison of elastic terms: $\pi^2 b/(nL)$ gives $\pi^2 \cdot 100/(92 \cdot 0.34) \approx 31.6$ vs. $\pi^2 \cdot 100/(401 \cdot 0.34) \approx 7.2$, in units of $k_B T$; the entropic correction $\frac{3}{2} \ln(401/92) \approx 2.2$ is too small to flip the inequality.)